Lecture 8: Torsion

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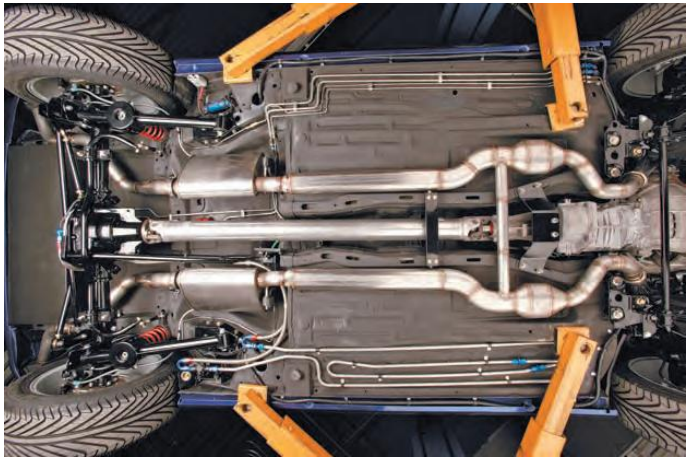
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Examples of Torsion Shafts



Power generation shaft



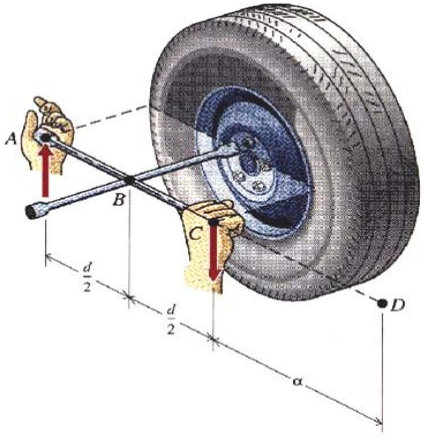
Automotive power train shaft



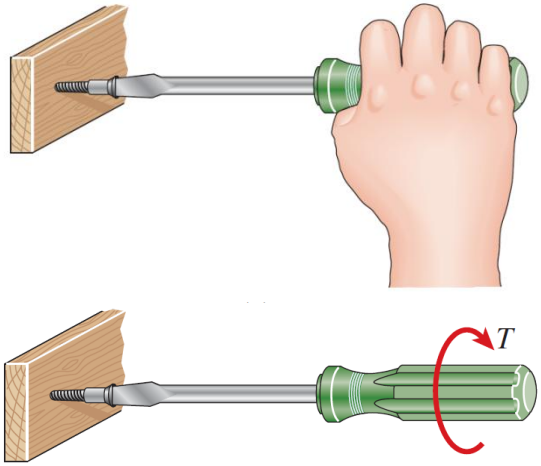
Ship drive shaft



Complex crank shaft

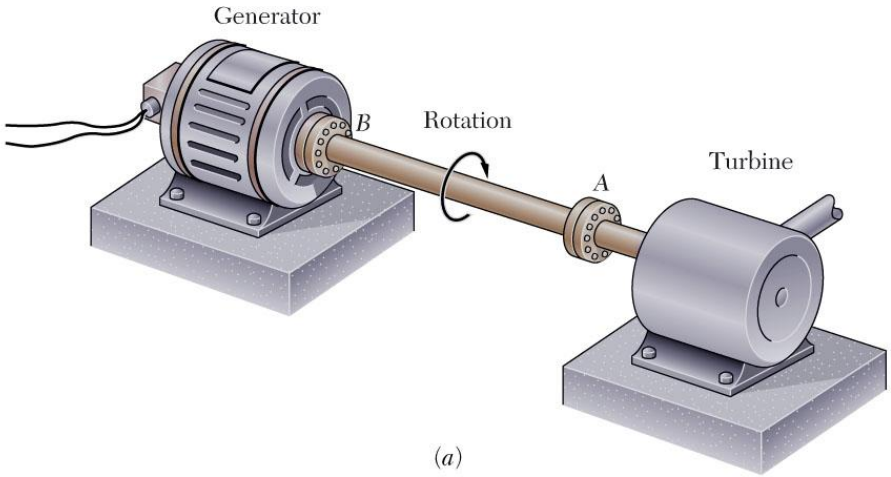


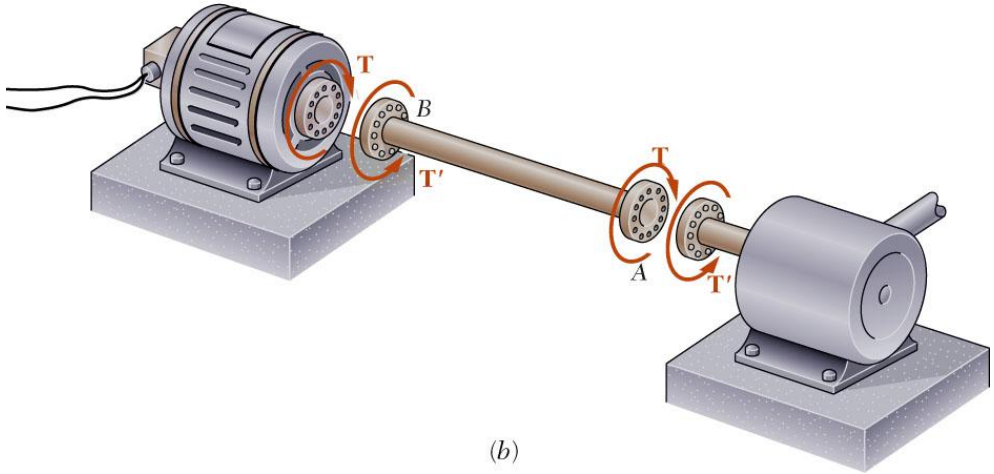
Tire shift drive



Screwdriver

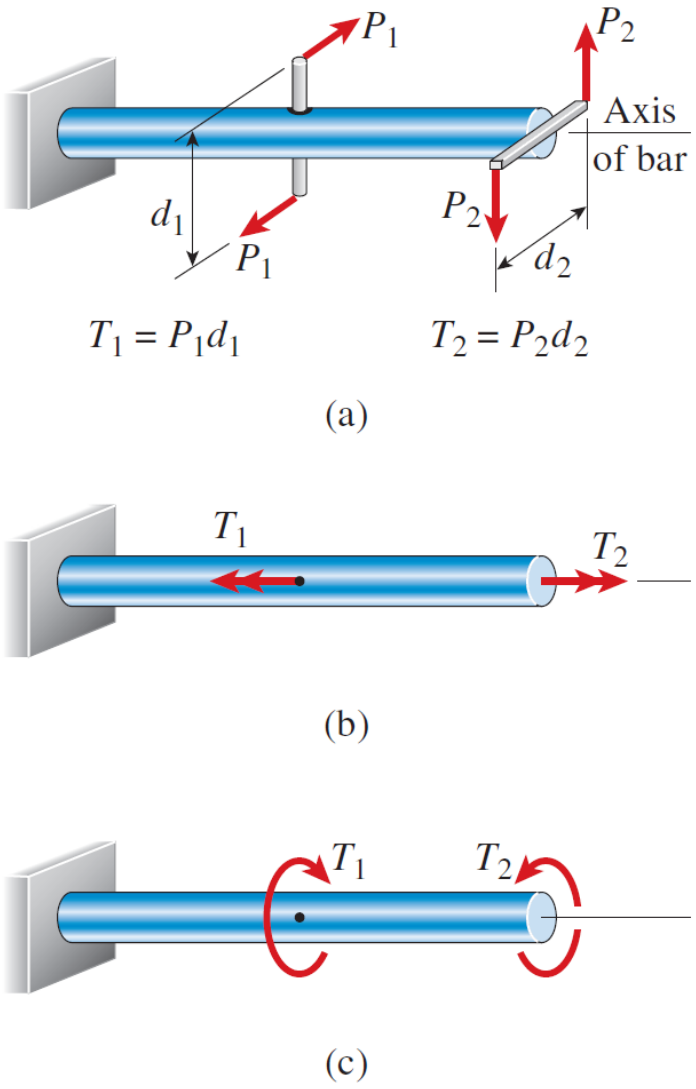
Introduction to Torsion





- Turbine exerts torque T on the shaft
- Shaft transmits the torque to the generator
- Generator creates an equal and opposite torque T
- Cross section remains planar
- Interested in stresses and strains of circular shafts subjected to twisting couples or torques

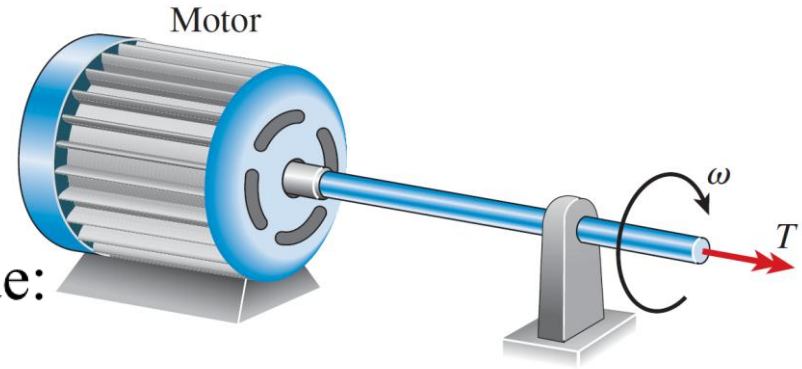
Circular bar subject to torsion by torques



- An idealized torsional loading is shown in left figure (a), the first pair are two forces of P_1 at distance d_1, and the second pair are two forces P_2 at distance d_2. Each pair of forces forms a **couple** that tends to twist the bar about its longitudinal axis.
- The moment of a couple equals the product of one force and the perpendicular distance between forces.
- The moment of a couple may be represented by a vector in the form of a double-headed arrow, the arrows are perpendicular to the plane containing the couple, and parallel to the axis of the bar. The direction of the moment is indicated by the right-hand rule for the moment vector. Use your right hand, let your fingers curl in the direction of the moment, and then your thumb will point in the direction of the vector.
- The torque (moment) can also be represented by a curved arrow acting in the direction of rotation.
- All these moments produce twisting of bar, are called **torques** or **twisting moments** or torsion moments

Power & Torque

- Transformation from electric power to mechanical power



- motor power: P (kW); generated torque: M_e (N·m): $W = \frac{J}{s} = \frac{Nm}{s}$
- n (rpm, revolutions per minute):

$$M_e \times 2\pi n = P \times 1000 \times 60 \Rightarrow M_e = \frac{1000 \times 60}{2\pi} \frac{P}{n} = 9549 \frac{P}{n}$$

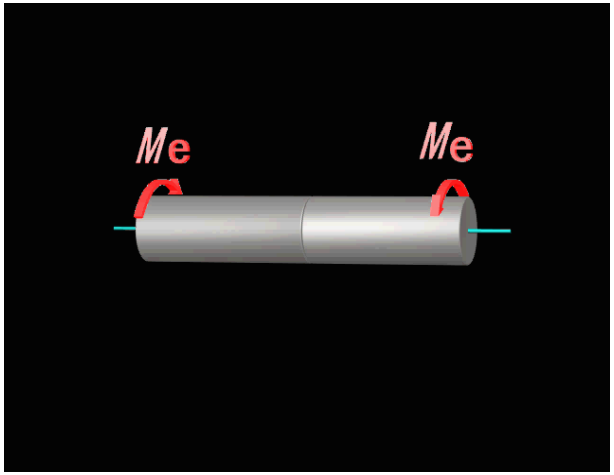
- n (rps, revolutions per second):

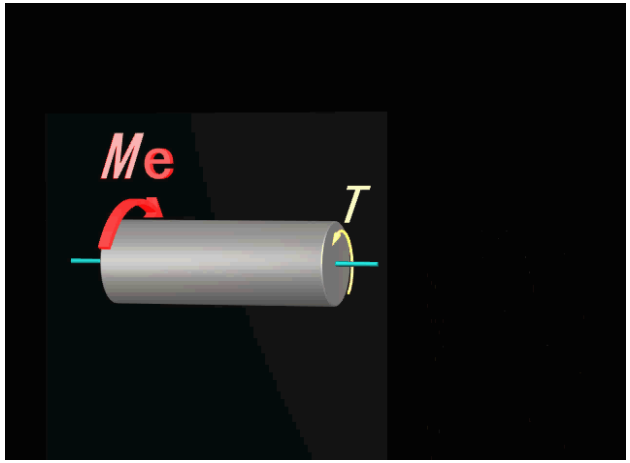
$$M_e \times 2\pi n = P \times 1000 \Rightarrow M_e = \frac{1000}{2\pi} \frac{P}{n} = 159 \frac{P}{n}$$

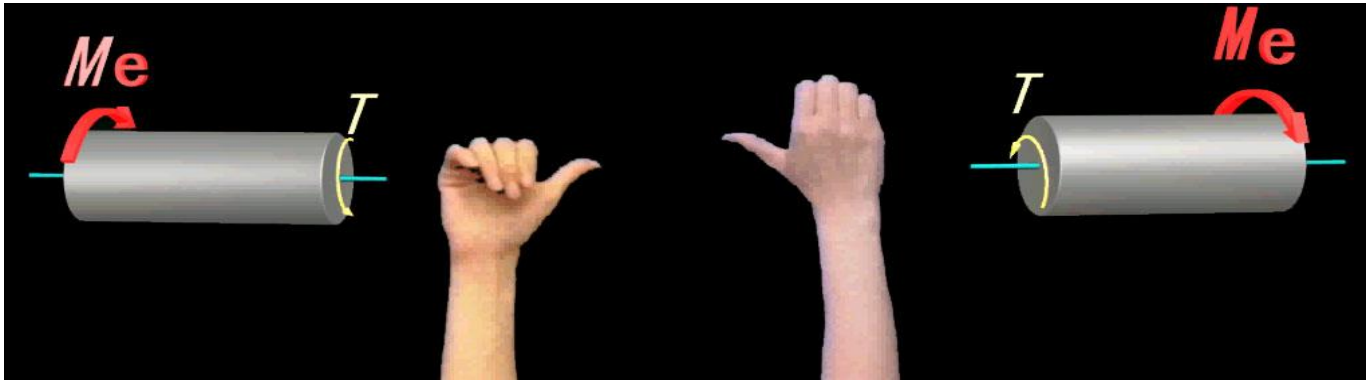
- ω (rad/s, radian per second):

$$M_e \times \omega = P \times 1000 \Rightarrow M_e = 1000 \frac{P}{\omega}$$

Sign Convention of Torque



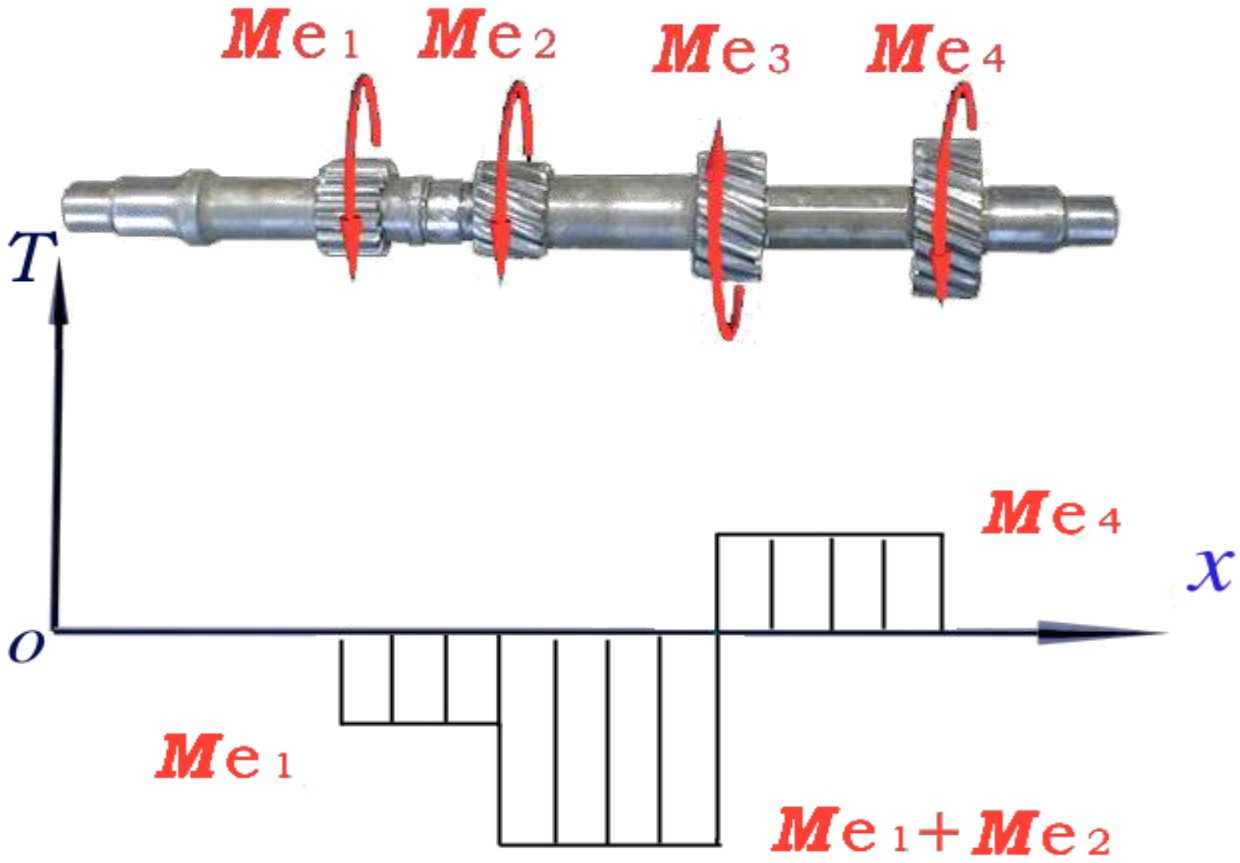




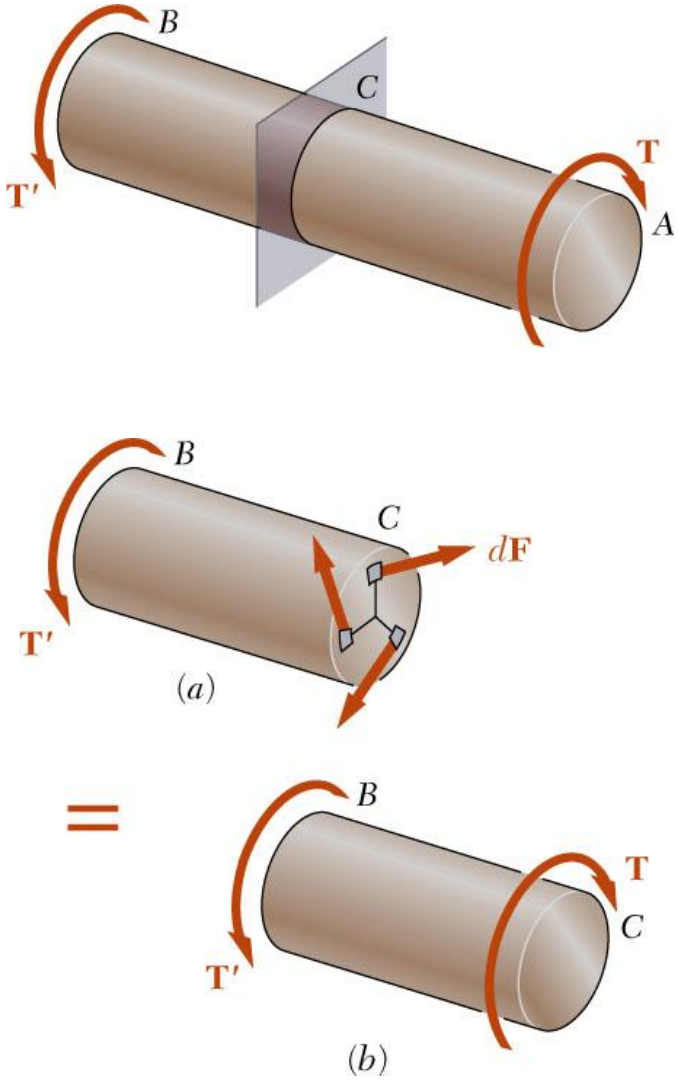
- Sign (Positive) convention (right-hand rule): thumb – cross-section normal; rest fingers – torque. This page shows positive torque because thumb points away from the material.

Torque Diagram

- Abscissa: cross-section position
- Ordinate: torque



Internal Torque & Stress - Static Indeterminacy



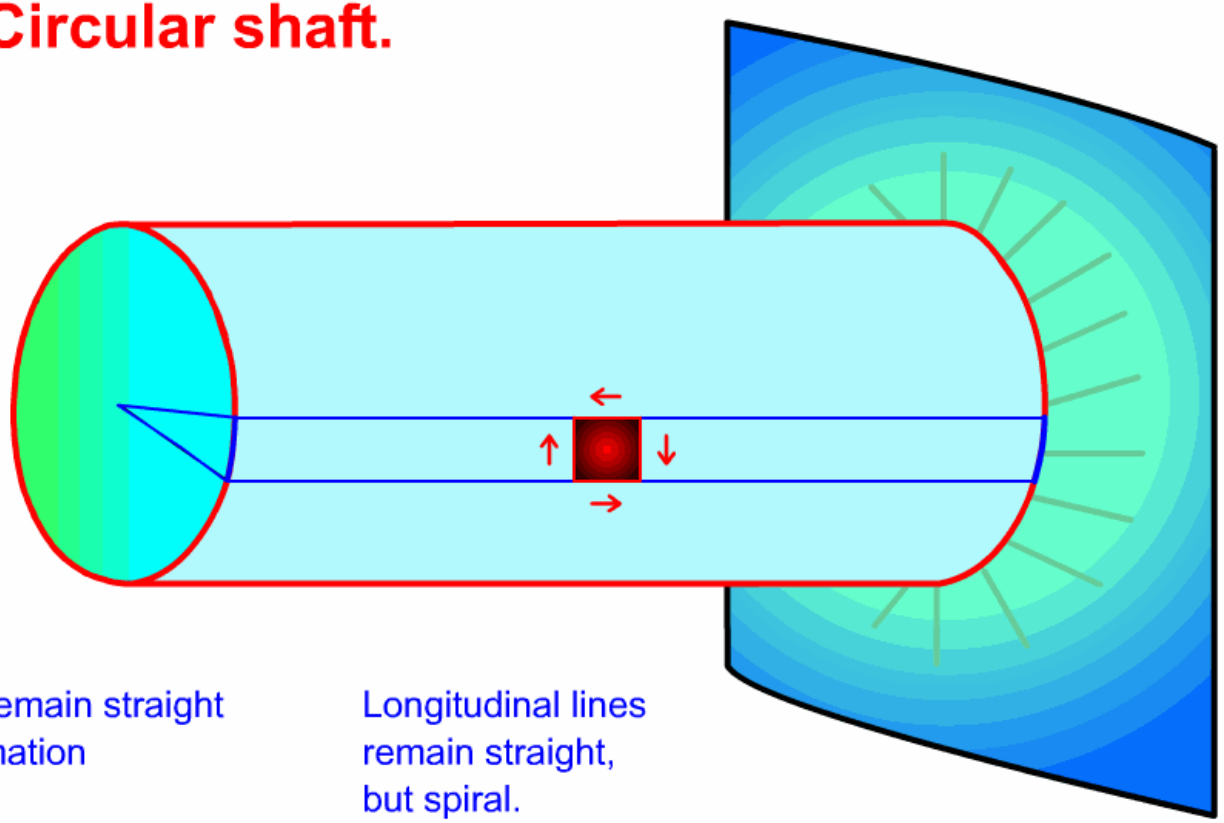
- Net of the internal shear stresses is an internal torque, equal and opposite to the applied torque,

$$T = \int \rho \, dF = \int \rho(\tau \, dA)$$

- Although the net torque due to the shear stresses is known, the distribution of the shear stresses is not.
- Distribution of shear stresses is statically indeterminate – must consider shaft deformations.
- Unlike the normal stress due to axial loads, the distribution of shear stresses due to torsional loads can not be assumed uniform.

Introduction to Torsion

Torsional Deformation of a Circular shaft.

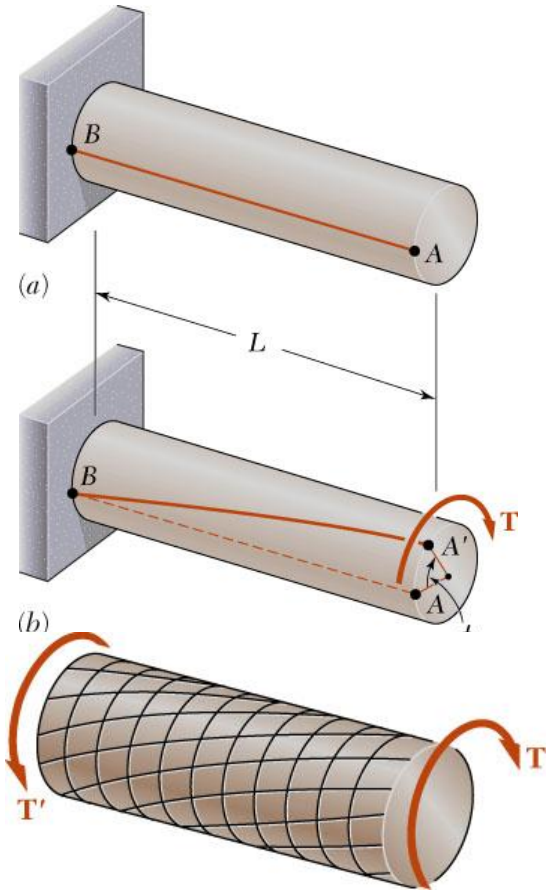


Radial lines remain straight during deformation

Longitudinal lines remain straight, but spiral.

Notice the deformation of the rectangular element when it is subjected to a torque

Kinematics



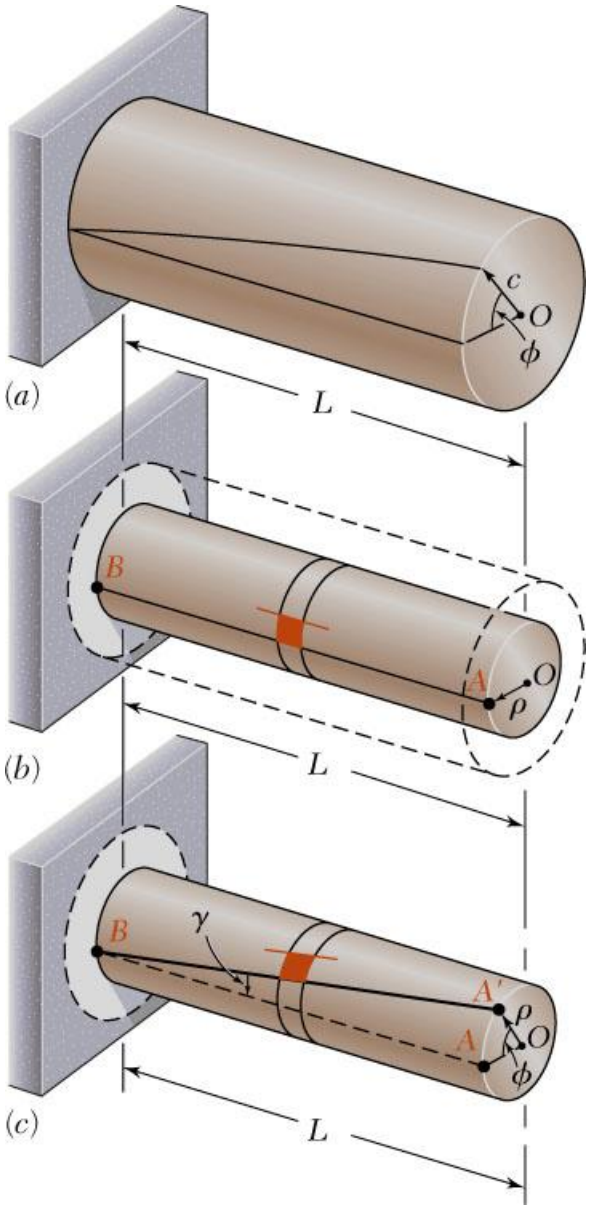
- From observation, the angle of twist of the shaft is proportional to the applied torque and to the shaft length. ϕ is the angle of twist at section A.

$$\phi \propto T, \qquad \phi \propto L$$

- When subjected to torsion, every cross-section of a circular shaft (hollow and solid) remains plane and undistorted. Because a circular shaft is axisymmetric.

- Since every cross section of the bar is identical, and since every cross section is subjected to the same internal torque, we say that the bar is in ***pure torsion***.

Kinematics



- Consider an interior section of the shaft. As a torsional load is applied, an element on the interior cylinder deforms into a rhombus.
- Since the ends of the element remain planar, the shear strain may be related to the angle of twist.
- It follows that

$$L\gamma = \rho\phi \quad \text{or} \quad \gamma = \frac{\rho\phi}{L}$$

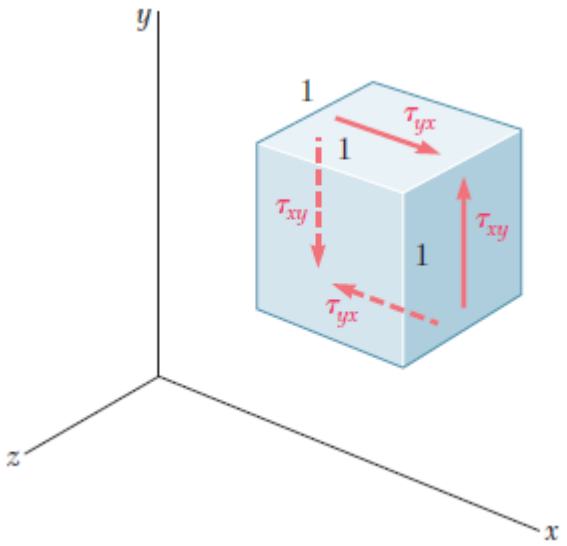
- Shear strain is proportional to the angle of twist and radius

c is the radius of the bar

ρ is radius at any point

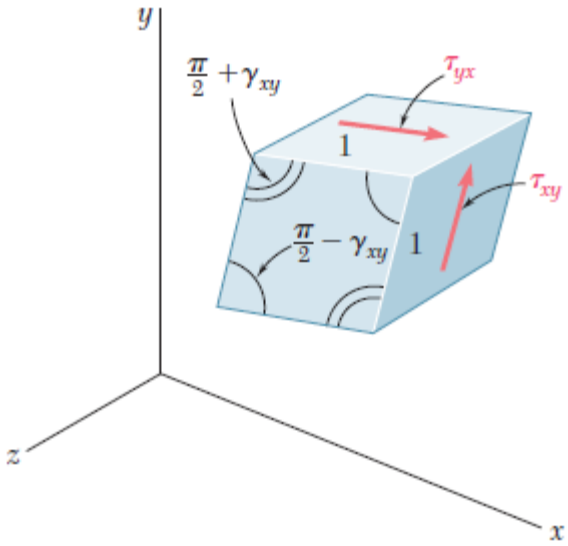
$$\gamma_{\max} = \frac{c\phi}{L} \quad \text{and} \quad \gamma = \frac{\rho}{c}\gamma_{\max}$$

Hooke's Law for Shear Deformation



- A cubic element subjected to a shear stress will deform into a rhomboid. The corresponding *shear* strain is quantified in terms of the change in angle between the sides,

$$\tau_{xy} = f(\gamma_{xy})$$

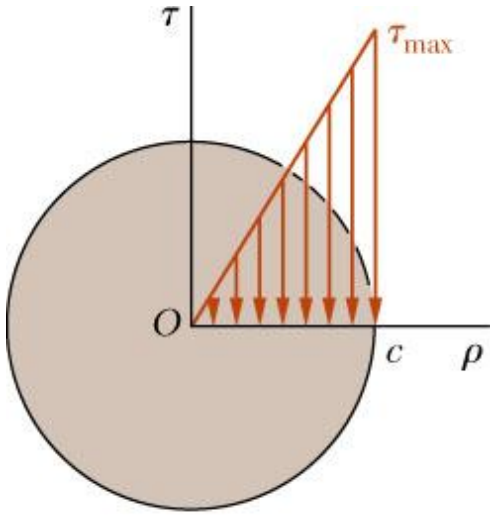


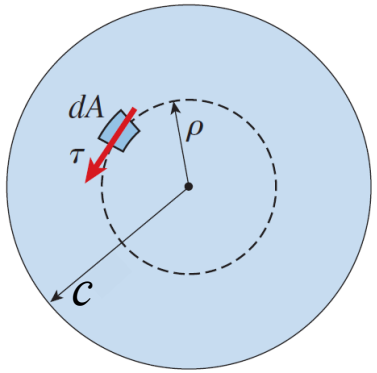
- For small strains

$$\tau_{xy} = G\gamma_{xy} \quad \tau_{yz} = G\gamma_{yz} \quad \tau_{zx} = G\gamma_{zx}$$

where G is the modulus of rigidity or shear modulus.

Static Equivalency





- At interior section of the shaft, shear strain

$$\gamma = \frac{\rho}{c}\gamma_{\max}$$

From Hooke's Law, $\tau = G\gamma$, so $\tau = \frac{\rho}{c}\tau_{\max}$

The shear stress varies linearly with the radial position in the section.

- Recall that the sum of the moments from the internal stress distribution is equal to the torque on the shaft at the section,

$$T = \int \rho\tau \ dA = \frac{\tau_{\max}}{c}\int \rho^2 \ dA = \frac{\tau_{\max}}{c} I_p$$

- The results are known as the *elastic torsion formulas*, (**I_p – Polar moment of inertia**) $\tau_{\max} = \frac{Tc}{I_p}$ and $\tau = \frac{T\rho}{I_p}$

Torsional Stress & Angle of Twist

- Torsional section modulus: [m^3]: $W_p = I_P / c$.
- Torsional stress

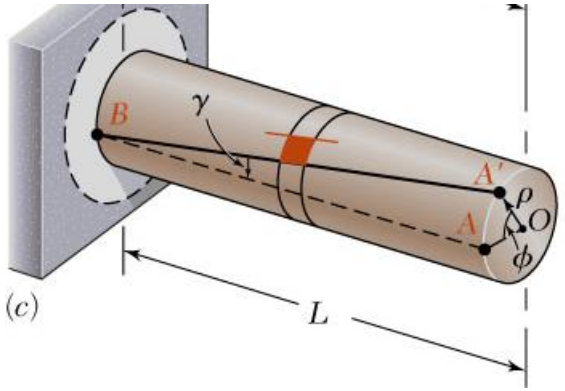
$$\Rightarrow \tau_{\max} = \frac{Tc}{I_p} = \frac{T}{W_p}, \qquad \tau = \frac{T\rho}{I_p} = \frac{T}{W_p}\frac{\rho}{c}$$

- Angle of twist ϕ: this equation is in radians, not degree

$$L\gamma = \rho\phi \Rightarrow \frac{\phi}{L} = \frac{\gamma}{\rho} = \frac{1}{\rho}\frac{\tau}{G} = \frac{1}{\rho}\frac{1}{G}\frac{T\rho}{I_p} = \frac{T}{GI_p} \Rightarrow \phi = \frac{TL}{GI_p}$$

- GI_p: Torsional rigidity [N•m^2]

- The equations are limited to bars of circular cross section (either solid or hollow) that behave in a linearly elastic manner.



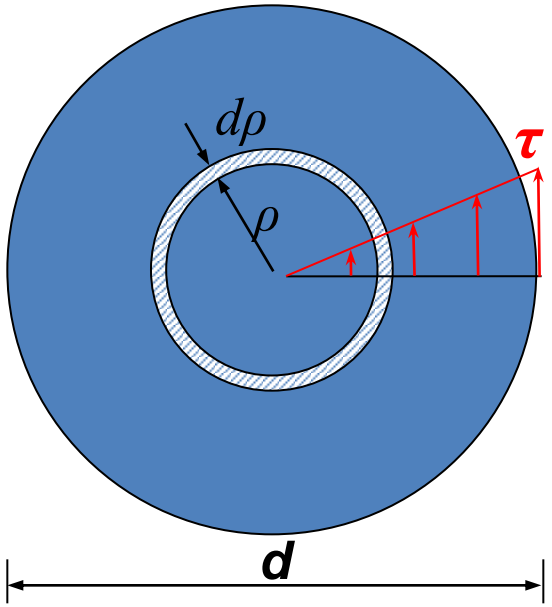
Polar Moment of Inertia & Section Modulus

- Solid circular shafts: *d* is diameter of the section

$$dA = 2\pi\rho d\rho$$

$$I_p = \int_A \rho^2 dA$$

$$= \int_0^{\frac{d}{2}} \rho^2 \cdot 2\pi\rho d\rho = \frac{\pi d^4}{32}$$



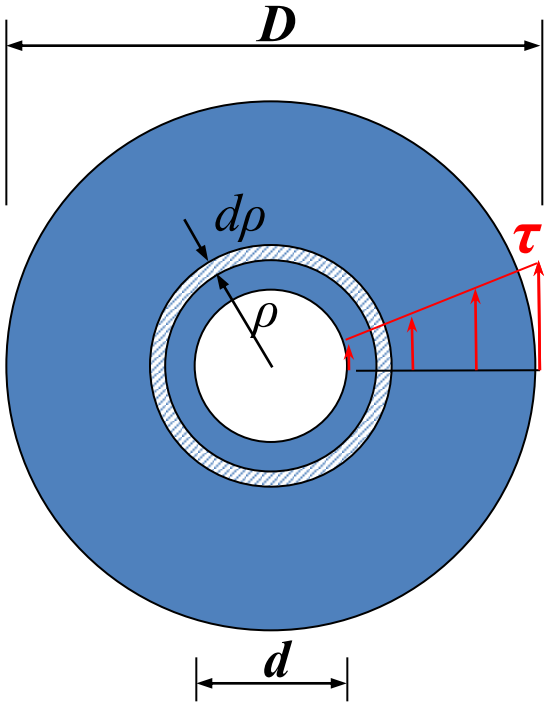
$$I_p = \int_A \rho^2 dA$$

$$= \int_A (y^2 + z^2)\, dA = I_y + I_z$$

$$W_p = \frac{I_p}{r} = \frac{\pi d^4}{32} \times \frac{2}{d} = \frac{\pi d^3}{16}$$

Polar Moment of Inertia & Section Modulus

- Hollow circular shafts: D is outer diameter, d is inner diameter



$$I_p = \int_A \rho^2 dA$$

$$= \int_{\frac{d}{2}}^{\frac{D}{2}} \rho^2 \cdot 2\pi\rho d\rho$$

$$= \frac{\pi}{32}\left(D^4 - d^4\right)$$

$$= \frac{\pi D^4}{32}\left(1-\alpha^4\right)$$

$$\left(\alpha = \frac{d}{D}\right)$$

$$W_p = \frac{I_p}{D/2}$$

$$= \frac{\pi D^3}{16}\left(1-\alpha^4\right)$$

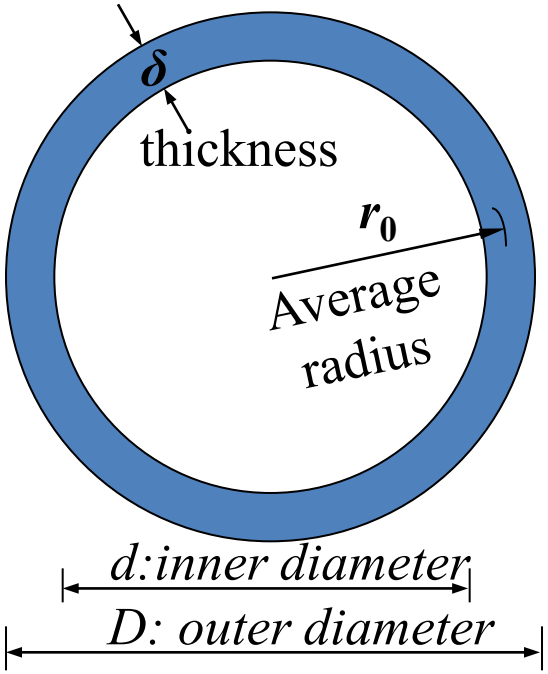
Polar Moment of Inertia & Section Modulus

- Thin-walled tubes: ($\delta < 0.1\, r_0$)

$$D = 2r_0 + \delta, \quad d = 2r_0 - \delta, \quad A \approx 2\pi r_0 \delta$$

$$I_p = \int_{\frac{d}{2}}^{\frac{D}{2}} \rho^2 \cdot dA \approx r_0^2 \int_{\frac{d}{2}}^{\frac{D}{2}} dA \approx 2\pi r_0^3 \delta$$

$$W_p = \frac{I_p}{r_0} \approx 2\pi r_0^2 \delta$$



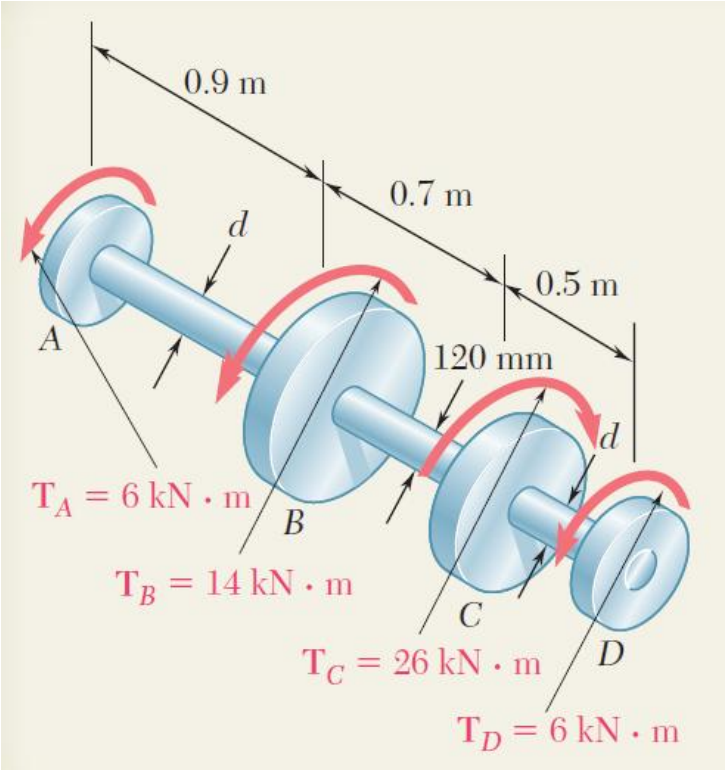
- **Torsional stress in thin-walled tubes can be simplified to**

$$\tau = \frac{T}{W_p} = \frac{T}{2\pi r_0^2 \delta}$$

- **In thin-walled tubes, we can assume shear stress is uniform at all points.**

$$T = \int_A r(\tau dA) = r_0 \tau \int_A dA = r_0 \tau 2\pi r_0 \delta \quad \Rightarrow \quad \tau = \frac{T}{2\pi r_0^2 \delta}$$

Example 6.1



Shaft *BC* is hollow with inner and outer diameters of 90 mm and 120 mm, respectively. Shafts *AB* and *CD* are solid of diameter *d*. For the loading shown, determine (*a*) the minimum and maximum shear stress in shaft *BC*, (*b*) the required diameter *d* of shafts *AB* and *CD* if the allowable shear stress in these shafts is 65 MPa.

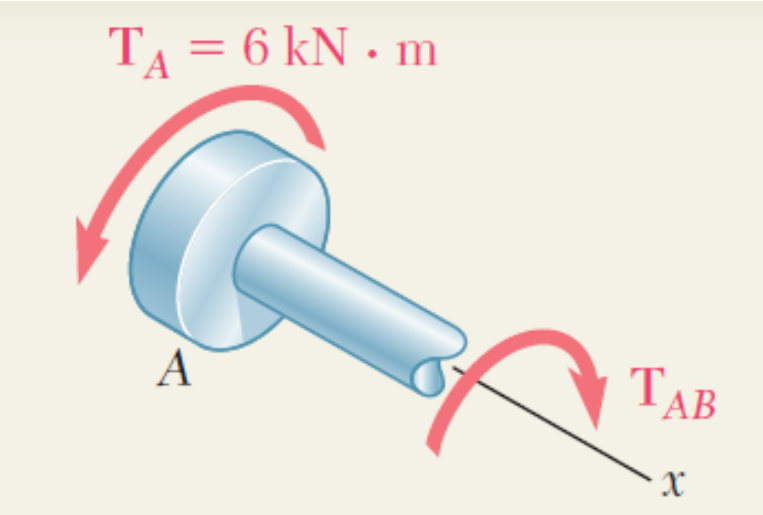
SOLUTION:

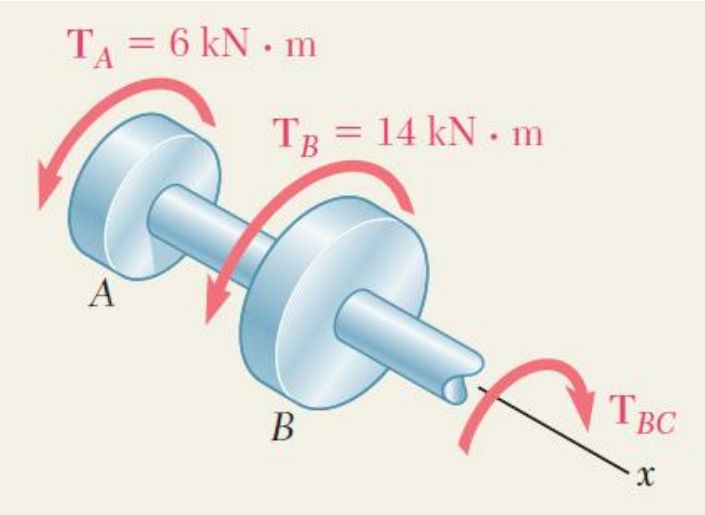
- Cut sections through shafts *AB* and *BC* and perform static equilibrium analysis to find torque loadings
- Apply elastic torsion formulas to find minimum and maximum stress on shaft *BC*
- Given allowable shear stress and applied torque, invert the elastic torsion formula to find the required diameter

Example 6.1

SOLUTION:

- Cut sections through shafts *AB* and *BC* and perform static equilibrium analysis to find torque loadings





$$\sum T = 0 = (6\,\text{kN}\cdot\text{m}) - T_{AB}$$

$$T_{AB} = 6\,\text{kN}\cdot\text{m} = T_{CD}$$

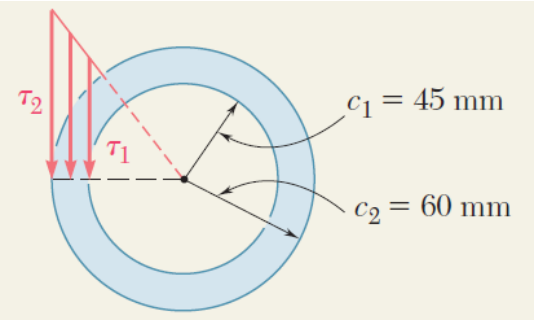
$$\sum T = 0 = (6\,\text{kN}\cdot\text{m}) + (14\,\text{kN}\cdot\text{m}) - T_{BC}$$

$$T_{BC} = 20\,\text{kN}\cdot\text{m}$$

Note: In this problem, the sign of torque does not matter. If you calculate angle of twist, the sign of torque needs to be clear (AB and BC are negative torque, CD in positive torque).

Example 6.1

- Apply elastic torsion formulas to find minimum and maximum stress on shaft *BC*



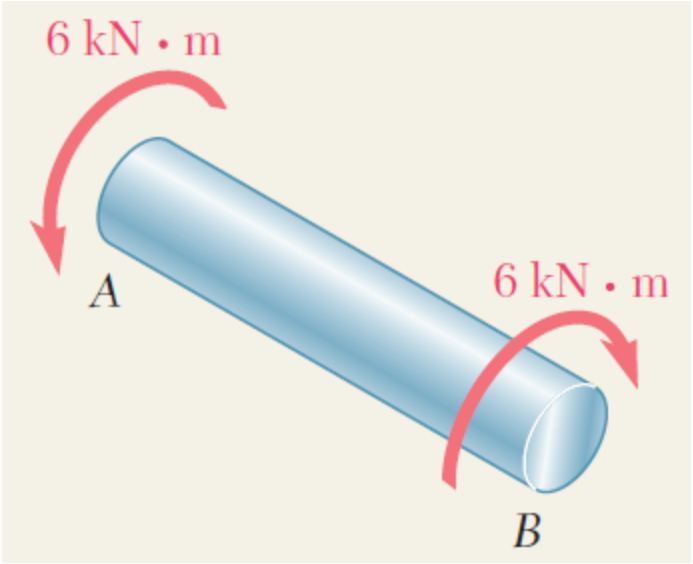
$$I_p = \frac{\pi}{2}\left(c_2^4 - c_1^4\right) = \frac{\pi}{2}\left[(0.060)^4 - (0.045)^4\right]$$
$$= 13.92\times10^{-6}\,\text{m}^4$$

$$\tau_{\max} = \tau_2 = \frac{T_{BC}c_2}{I_p} = \frac{(20\,\text{kN}\cdot\text{m})(0.060\,\text{m})}{13.92\times10^{-6}\,\text{m}^4}$$
$$= 86.2\,\text{MPa}$$

$$\frac{\tau_{\min}}{\tau_{\max}} = \frac{c_1}{c_2} \qquad \frac{\tau_{\min}}{86.2\,\text{MPa}} = \frac{45\,\text{mm}}{60\,\text{mm}}$$

$$\tau_{\min} = 64.7\,\text{MPa}$$

- Given allowable shear stress and applied torque, invert the elastic torsion formula to find the required diameter

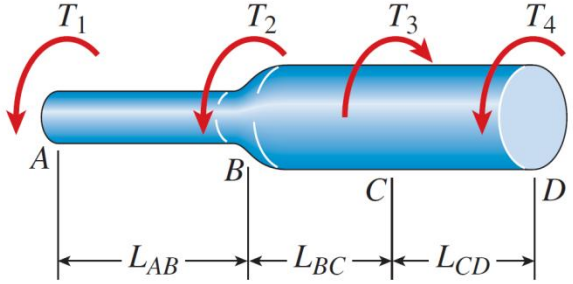


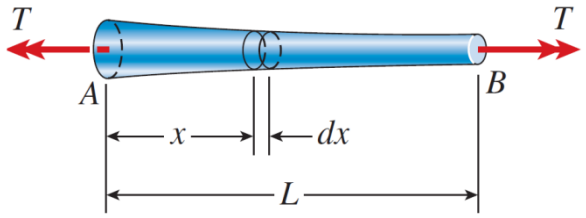
$$\tau_{\max} = \frac{Tc}{I_p} = \frac{Tc}{\frac{\pi}{2}c^4} \qquad 65MPa = \frac{6\,\text{kN}\cdot\text{m}}{\frac{\pi}{2}c^3}$$

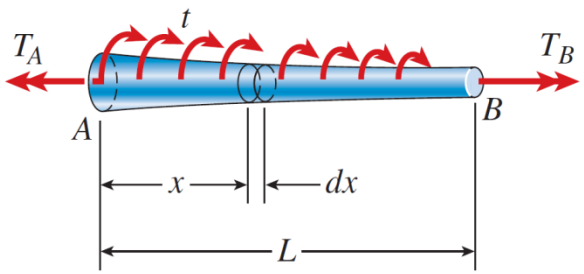
$$c = 38.9\times10^{-3}\,\text{m}$$
$$d = 2c = 77.8\,\text{mm}$$

Nonuniform Torsion of Circular Shafts

- Pure torsion formula refers to torsion of a prismatic bar subjected to torques acting only at the ends.
- Nonuniform torsion differs from pure torsion in that the bar need not to be prismatic and the applied torques may act anywhere along the axis of the bar.
- Bars in nonuniform torsion can be analyzed by applying the formulas of pure torsion to finite segments of the bar and then adding the results, or by applying the formulas to differential elements of the bar and then integrating.





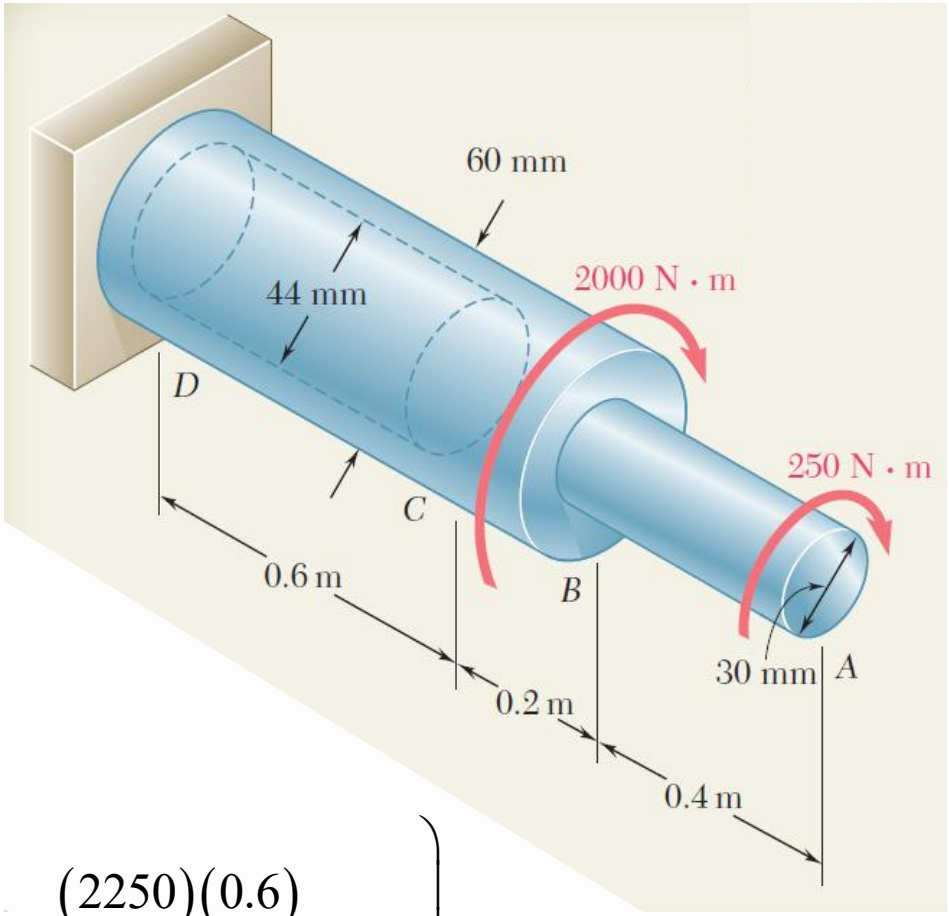


$$\phi = \sum_i \phi_i = \sum_i \frac{T_i L_i}{G_i \left(I_p\right)_i}$$

$$\phi = \int_0^L d\phi = \int_0^L \frac{T(x)\,dx}{G I_p(x)}$$

Example 6.2

- The horizontal shaft *AD* is attached to a fixed base at *D* and is subjected to the torques shown. A 44-mm-diameter hole has been drilled into portion *CD* of the shaft. Knowing that the entire shaft is made of steel for which $G = 77$ GPa, determine the angle of twist at end *A*.



SOLUTION:

$$\phi = \frac{1}{G}\sum_i \left(\frac{T_{AB}L_{AB}}{I_{pAB}} + \frac{T_{BC}L_{BC}}{I_{pBC}} + \frac{T_{CD}L_{CD}}{I_{pCD}} \right)$$

$$= \frac{1}{G}\sum_i \left(\frac{(250)(0.4)}{\frac{\pi}{2}(0.015)^4} + \frac{(2250)(0.2)}{\frac{\pi}{2}(0.03)^4} + \frac{(2250)(0.6)}{\frac{\pi}{2}\left[(0.030)^4 - (0.022)^4\right]} \right)$$

$$= 0.01634 + 0.00459 + 0.01939$$

$$= 0.0403 \text{ rad} = 0.0403(360/2\pi) = 2.31^0$$

$$\phi = \frac{TL}{GI_p}$$

Strength Analysis

- Strength check: $\tau_{\max} = (T/W_P)_{\max} \leq [\tau]$

- Cross-section design: $W_p \geq T_{\max}/[\tau]$

- Maximum allowable load: $T_{\max} \leq W_p[\tau]$

Stiffness Condition

- Angle of twist per unit length: $\phi' = \phi / L = T / GI_p$

$$\phi = \frac{TL}{GI_p}$$

- Stiffness condition: $\phi'_{\max} \le [\phi']$

  $[\phi']$: allowable angle of twist per unit length

  [radian/m] or [degrees/m]

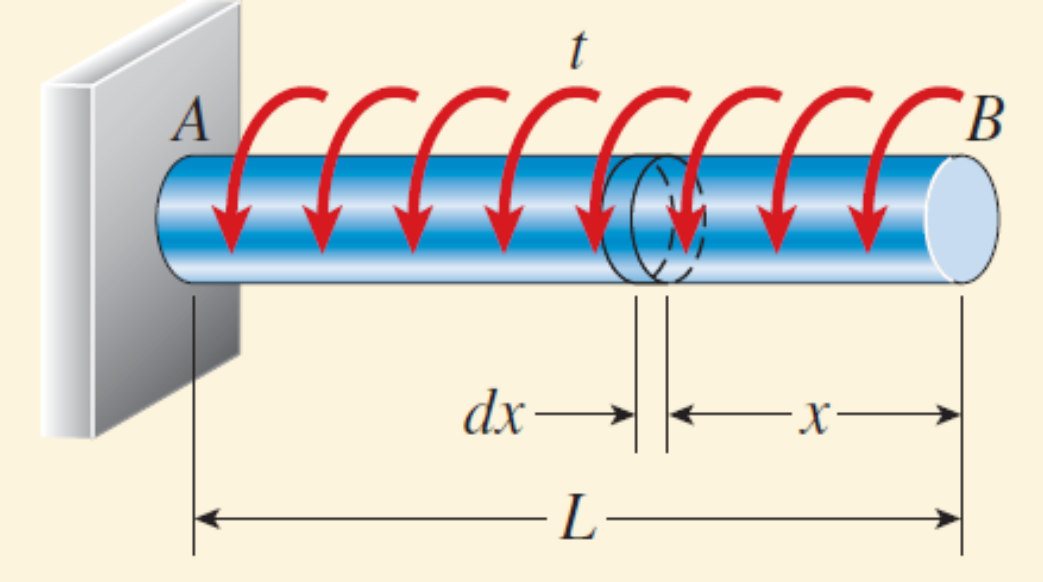
  ✓NC machine: $[\phi']$ = 0.15-0.3 degrees/m

  ✓Ordinary shaft: $[\phi']$ = 0.5-2.0 degrees/m

  ✓Shaft of drilling machine: $[\phi']$ = 2.0-4.0 degrees/m

- Stiffness check;
- Cross-section design;
- Allowable load.

Example 6.3



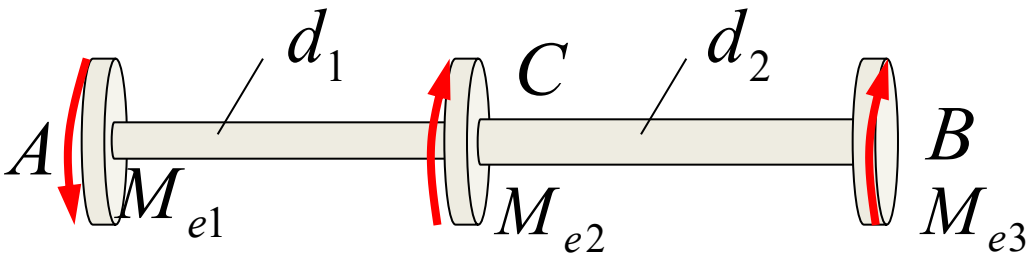
- Determine the angle of twist at cross-section *B* of the shaft. In this example, uniformly distributed torque (*t*) is applied along the length.

- Solution

$$\phi_B = \int_0^L \frac{T}{GI_p}\,dx = \int_0^L \frac{tx}{GI_p}\,dx = \frac{tL^2}{2GI_p}$$

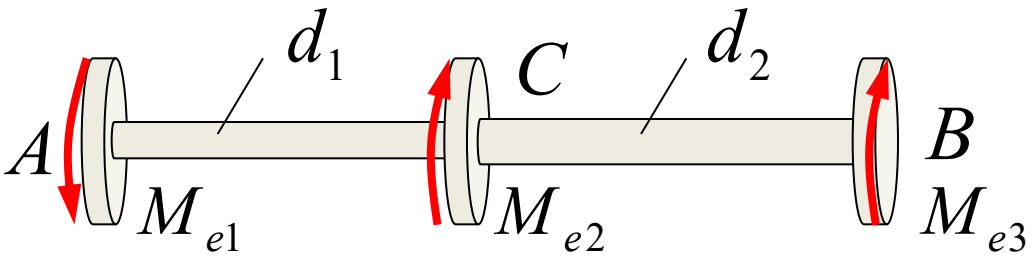
Example 6.4

- Given: n = 500 rpm, input gear A (P_1 = 400 kW), output gears C and B (P_2 = 160 kW, P_3 = 240 kW), $[\tau]$ = 70 MPa, $[\phi']$ = 1 degree/m (allowable angle of twist per length), G = 80 GPa. Find (1) d_1 and d_2; (2) d if $d_{AC} = d_{BC}$ is required; (3) How to arrange the input and output gears to make the transmission more reasonable?

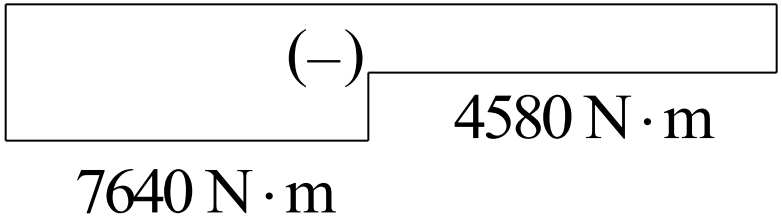


- Solution:

1. Torque diagram



$$M_{e1} = 9549\frac{P_1}{n} = 9549 \times \frac{400}{500}$$
$$= 7640\ \mathrm{N \cdot m}$$



$$M_{e2} = \frac{160}{400} M_{e1} = 3060\ \mathrm{N \cdot m}$$

$$M_{e3} = \frac{240}{400} M_{e1} = 4580\ \mathrm{N \cdot m}$$

Note: in this problem, n = 500 rpm is correct for both AB and BC.

2. Find d_1

- From strength condition:

$$\tau_{\max} = \frac{T}{\pi d_1^3/16} \le [\tau]$$

$$\Rightarrow d_1 \ge \sqrt[3]{\frac{16T}{\pi[\tau]}} = \sqrt[3]{\frac{16\times 7640}{\pi\times 70\times 10^6}} = 82.2\times 10^{-3}\ \text{m} = 82.2\ \text{mm}$$

- From stiffness condition:

$$\phi'_{\max} = \frac{T}{G\pi d_1^4/32}\times\frac{180^\circ}{\pi} \le [\phi']$$

$$\Rightarrow d_1 \ge \sqrt[4]{\frac{32T\times 180}{G\pi^2\times[\phi']}} = \sqrt[4]{\frac{32\times 7640\times 180}{80\times 10^9\times\pi^2\times 1}} = 86.4\times 10^{-3}\ \text{m} = 86.4\ \text{mm}$$

- Take $d_1 = 86.4$ mm

3. Find d_2

- From strength condition:

$$d_2 \geq \sqrt[3]{\frac{16T}{\pi[\tau]}} = \sqrt[3]{\frac{16\times 4580}{\pi\times 70\times 10^6}} = 69.3\times 10^{-3}\,\mathrm{m} = 69.3\mathrm{mm}$$

- From stiffness condition:

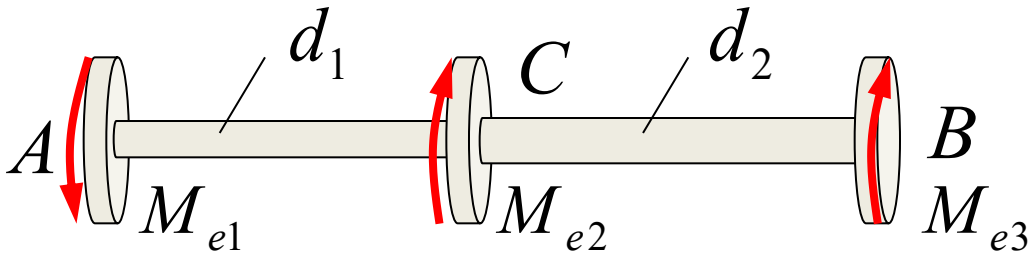
$$d_2 \geq \sqrt[4]{\frac{32T\times 180}{G\pi^2\times[\phi']}} = \sqrt[4]{\frac{32\times 4580\times 180}{80\times 10^9\times\pi^2\times 1}} = 76\times 10^{-3}\,\mathrm{m} = 76\mathrm{mm}$$

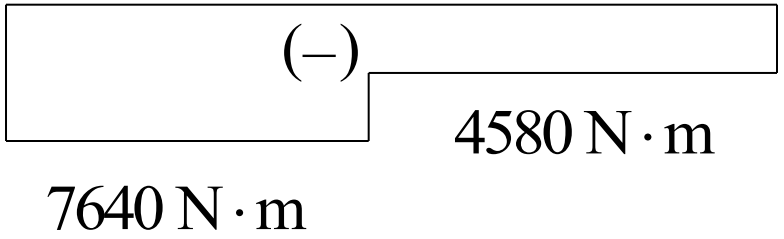
- Take $d_2 = 76\mathrm{mm}$

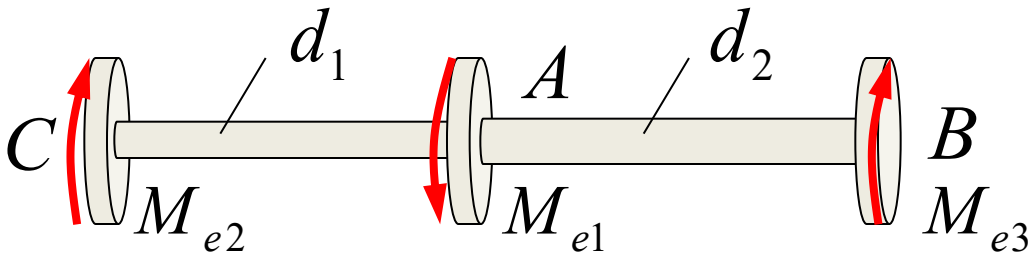
4. If $d = d_1 = d_2$ is required, take the maximum value:

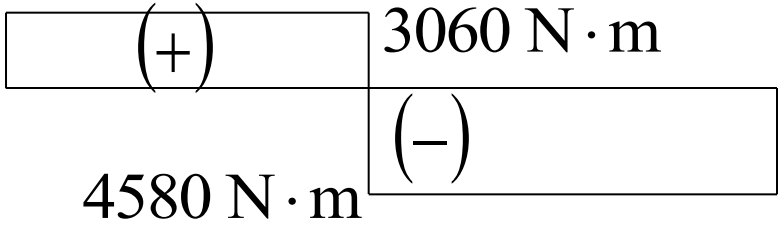
$d = d_1 = 86.4\mathrm{mm}$

5. Rearrangement of gears to make the transmission more efficient.
   Put the input gear in between two output gears.







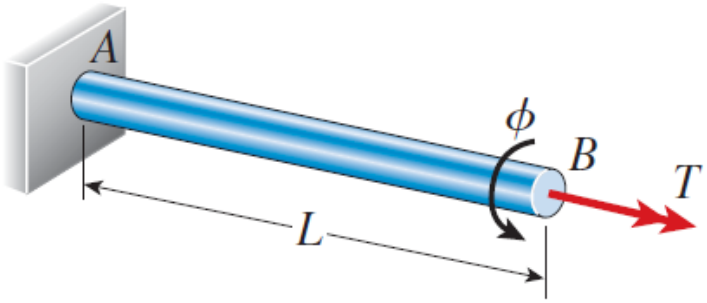


- Smaller maximum load carried by the system (4580/7640=60%)

Strain Energy=External work done by load *T*

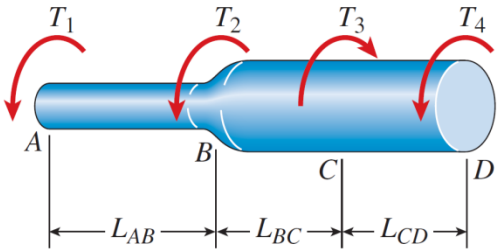
- Strain energy and work in pure torsion

$$U = W = \frac{1}{2}T\phi = \frac{T^2 L}{2GI_p} = \frac{GI_p}{2L}\phi^2 \qquad \left(\phi = \frac{TL}{GI_p}\right)$$

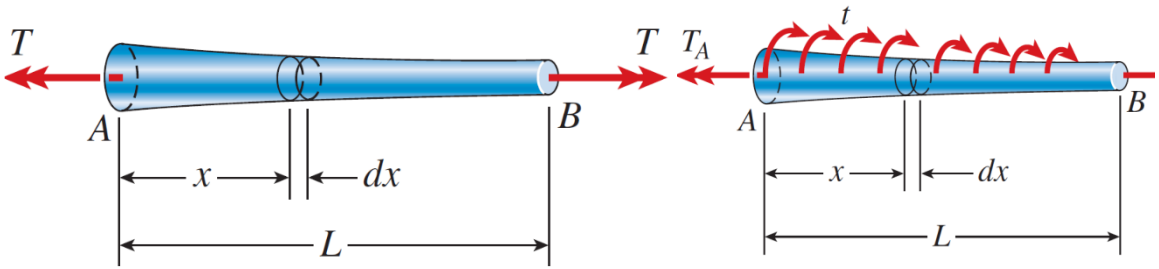


Prismatic bar in pure shear

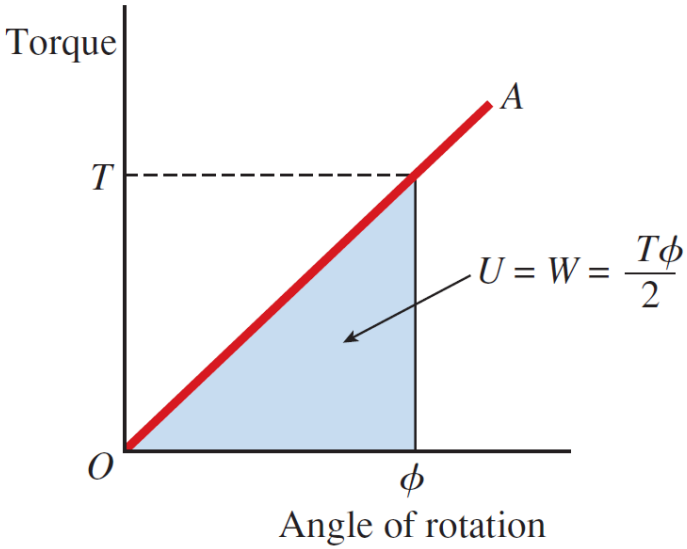
- Nonuniform torsion



$$U = \sum_i U_i = \sum_i \frac{T_i^2 L_i}{2G_i\left(I_p\right)_i}$$



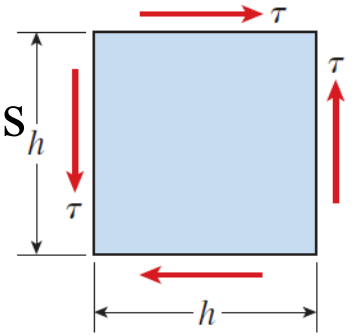
$$U = \int_0^L dU = \int_0^L \frac{\left[T(x)\right]^2 dx}{2GI_p(x)}$$



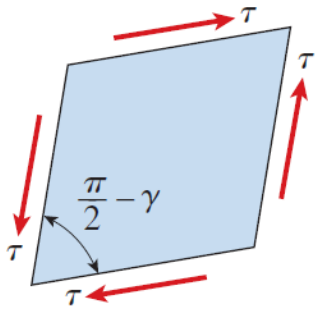
Torque-rotation diagram

Strain Energy Density

- Consider a differential cube of side h subjected to shear stresses τ on its sides



(a)

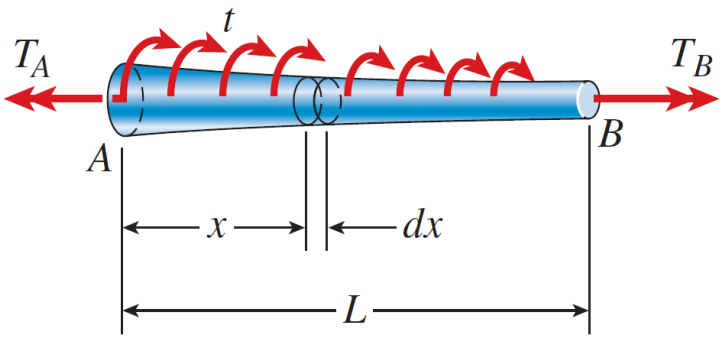


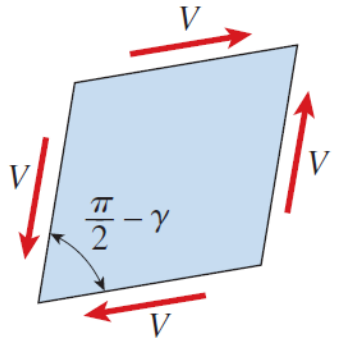
(b)

$$U = W = \frac{1}{2} V\delta = \frac{1}{2}\left(\tau h^2\right)\left(\gamma h\right) = \frac{1}{2}\tau\gamma h^3$$

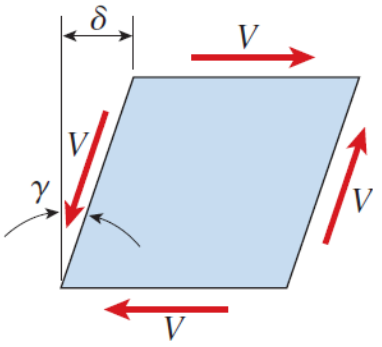
$$u = \frac{U}{h^3} = \frac{1}{2}\tau\gamma = \frac{1}{2}G\gamma^2 = \frac{1}{2G}\tau^2$$

- Total strain energy in torsion





(c)



(d)

$$U = \int_V u dV = \int_L \int_A u dA dx = \int_L \int_A \frac{\tau^2}{2G} dA dx$$

$$= \int_L \int_A \frac{1}{2G}\left(\frac{T\rho}{I_p}\right)^2 dA dx = \int_L \frac{T^2}{2GI_p^2}\left\{\int_A \rho^2 dA\right\} dx = \int_L \frac{T^2 dx}{2GI_p}$$

Important equation to remember

Angle of twist per unit length θ under pure torsion:

$$\theta = \frac{T}{GI_P}$$

Total Angle of twist $\phi = \theta$ of bar under pure torsion:

$$\phi = \frac{TL}{GI_P}$$

Torsional stiffness of bar (defined by T divided by total angle of twist ϕ):

$$k_T = \frac{GI_P}{L}$$

Shear stress at distance ρ from center of bar (c is radius of bar, or outer radius of cylinder):

$$\tau = \frac{\rho}{c}\tau_{\max} = \frac{T\rho}{I_P}$$